Wenqin Chen

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Research Interests

Machine learning for quantum many-body physics (neural quantum states); quantum geometry of superconductivity and phonon magnetism; topological phases, skyrmion crystals, and quantum anomalous Hall states; topological quantum computing with Majorana zero modes.

Education

University of Washington, Seattle, WA — 2021 – 2027 (expected)
PhD in Physics. Advisors: Di Xiao, Ting Cao

- Thesis: Machine Learning and Quantum Many-Body Theory for Condensed Matter Physics

Xi'an Jiaotong University, Xi'an, China — 2016 – 2021
BS in Physics (Honors)

- Outstanding Undergraduate Thesis Award

University of California, Berkeley, Berkeley, CA — 2019
Exchange student in Physics

Research Experience

University of Washington, Seattle, WA — 2021 – Present
Graduate Research Assistant. Advisors: Di Xiao, Ting Cao

- Neural quantum states: built a transformer wavefunction ansatz that solves the continuum many-body Schrödinger equation in JAX, optimized by variational Monte Carlo with natural-gradient (stochastic-reconfiguration) updates, warm-started from Hartree-Fock and benchmarked against exact diagonalization; trained on H200 GPUs (UW Tillicum) and Azure cloud.
- Quantum geometry of superconductivity: developed the Cooper-pair quadrupole framework showing Berry curvature and the quantum metric set a lower bound on the pair size in flat-band superconductors [arXiv:2605.03133].
- Phonon magnetism: built gauge-theory and geometric descriptions of giant phonon magnetic moments in Dirac materials [PRB 111, 035126; arXiv:2505.09732; arXiv:2605.06983], with predictions guiding magneto-Raman and ultrafast experiments by collaborators at Los Alamos (CINT) and UC Irvine.
- Co-developed the theory of adiabatic pumping of orbital magnetization by spin precession [PRL 134, 176702].

Los Alamos National Laboratory (T-4 / CNLS), Los Alamos, NM — Summers 2023 & 2024
Research Intern. Mentor: Shi-Zeng Lin

- Built a self-consistent Hartree-Fock mean-field framework to search for skyrmion crystals and quantum anomalous Hall order in strongly correlated two-dimensional electron systems.
- Ran large-scale parameter-scan campaigns on the DOE NERSC Perlmutter supercomputer (SLURM, 128 cores/task); the research codebase (~28k lines, 148 tests) is maintained on GitHub.
- Delivered two first-author papers on phonon magnetism in consecutive summers [PRB 111, 035126; arXiv:2505.09732].

Lawrence Berkeley National Laboratory, Berkeley, CA — 2019
Research Intern. Mentor: Joseph Orenstein

- Developed numerical simulations to interpret optical birefringence measurements of the three-state nematic antiferromagnet Fe1/3NbS2; co-authored the resulting paper [Nature Materials (2020)].

Xi'an Jiaotong University, Xi'an, China — 2019 – 2021
Undergraduate Researcher. Mentor: Jie Liu

- Simulated non-Abelian braiding of Majorana zero modes in a tetron device and showed geometric-phase cancellation restores braiding-time-independent statistics in the presence of an Andreev bound state; first-author paper [PRB 105, 054507].
- Co-developed a minimal quantum-dot braiding protocol with electrical fermion-parity readout [SCPMA 64, 117811].

Publications

Nonadiabatic Theory of Phonon Magnetic Moments in Insulators and Metals — 2026
Haoran Chen, ***Wenqin Chen***, Kaijie Yang, Ting Cao, Di Xiao · *arXiv:2605.06983*

Quantum Geometric Quadrupole of Cooper Pairs — 2026
Wenqin Chen, Kaijie Yang, Ting Cao, Shi-Zeng Lin, Jiabin Yu, Di Xiao · *arXiv:2605.03133*

Geometric Origin of Phonon Magnetic Moment in Dirac Materials — 2025
Wenqin Chen, Xiao-Wei Zhang, Ting Cao, Shi-Zeng Lin, Di Xiao · *arXiv:2505.09732*

Adiabatic Pumping of Orbital Magnetization by Spin Precession — 2025
Yafei Ren, ***Wenqin Chen***, Chong Wang, Ting Cao, Di Xiao · *Physical Review Letters 134, 176702*

Gauge Theory of Giant Phonon Magnetic Moment in Doped Dirac Semimetals 2025
Wenqin Chen, Xiao-Wei Zhang, Ying Su, Ting Cao, Di Xiao, Shi-Zeng Lin · *Physical Review B 111, 035126*

Non-Abelian Statistics of Majorana Zero Modes in the Presence of an Andreev Bound State 2022
Wenqin Chen, Jiachen Wang, Yijia Wu, Jie Liu, X. C. Xie · *Physical Review B 105, 054507*

Minimal Setup for Non-Abelian Braiding of Majorana Zero Modes 2021
Jie Liu, ***Wenqin Chen***, Ming Gong, Yijia Wu, X. C. Xie · *Science China Physics, Mechanics & Astronomy 64, 117811*

Observation of Three-State Nematicity in the Triangular Lattice Antiferromagnet Fe1/3NbS2 2020
Arielle Little, Changmin Lee, Caolan John, Spencer Doyle, Eran Maniv, Nityan L. Nair, ***Wenqin Chen***, Dylan Rees, Jörn W. F. Venderbos, Rafael Fernandes, James G. Analytis, Joseph Orenstein · *Nature Materials 19, 1062*

Talks and Presentations

Quantum Geometric Quadrupole of Cooper Pairs in Flat-Band Superconductors — MEM·C IRG-2 Seminar, University of Washington *(invited seminar)*, Nov 2025

Gauge and Geometric Origins of Phonon Magnetism in Dirac Materials — Oka Group Seminar, Institute for Solid State Physics, University of Tokyo, online *(invited seminar)*, May 2025

Quantum Geometric Quadrupole of Cooper Pairs — APS Global Physics Summit, Denver *(contributed)*, Mar 2026

Geometric Theory of Phonon Magnetic Moment in Dirac Materials — APS Global Physics Summit, Anaheim *(contributed)*, Mar 2025

Theory of Giant Phonon Magnetic Moment in Doped Dirac Semimetals — APS March Meeting, Minneapolis *(contributed)*, Mar 2024

Phonon Magnetic Moments in Cd3As2: Theory Guiding the Magneto-Raman Experiment — presentations to the experimental collaboration, Los Alamos National Laboratory, online, Sep 2024

LANL–University of Michigan Joint Workshop on Quantum Materials — Los Alamos, NM *(poster)*, Jul 2025

Awards and Fellowships

- Clean Energy Institute Graduate Fellowship, University of Washington (2024–2025)
- Thouless Quantum Matter Graduate Fellowship, University of Washington
- Outstanding Undergraduate Thesis Award, Xi'an Jiaotong University (2021)

Technical Skills

Machine learning: Neural networks (transformer/attention architectures, neural quantum states), variational Monte Carlo, natural-gradient optimization (stochastic reconfiguration), training stabilization (gradient clipping, learning-rate schedules, warm starts)

Programming: Python (NumPy, SciPy, pandas), JAX (Flax, Optax), PyTorch, Bash/Linux, Git, pytest

Scientific computing: Electronic-structure methods (exact diagonalization, self-consistent-field Hartree-Fock), Bogoliubov-de Gennes / Nambu modeling of superconductor-semiconductor (Majorana) nanowire devices, Berry-curvature and transport (Kubo) calculations, stochastic methods, numerical linear algebra

Infrastructure: GPU training in JAX (H200), SLURM/HPC job orchestration (NERSC Perlmutter, UW Tillicum), Azure cloud compute, reproducible research pipelines

Service and Activities

- Organizer, condensed-matter-theory group meetings, University of Washington (2025 – Present)
- Co-organizer, Summer Research Symposium for condensed matter physics, University of Washington (2025)
- Member, American Physical Society (Division of Condensed Matter Physics)